SHADOWFOX DATA ANALYST INTERNSHIP
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Customer and Revenue Analysis

Online Retail — Intermediate Level
11 October 2026

EXECUTIVE SUMMARY

This analysis answers where revenue comes from, how well the retailer keeps its customers, and which actions are most likely to grow revenue. The cleaned dataset contains 522,537 completed product lines, 19,771 invoices, 4,333 identified customers, 3,908 products and 38 countries. Revenue is **£10.00M**; cancellations are £230K, or 2.3% of revenue.

Metric	Definition	Result
Revenue	Quantity multiplied by UnitPrice, completed sales	£10.00M
Average order value	Revenue divided by invoices	£505.84
Repeat customer rate	Customers with at least two orders	65.3%
Repeat revenue share	Revenue from repeat customers	93.5%
Top 10% concentration	Revenue from largest 10% of identified customers	60.3%
Identified revenue share	Revenue attached to a CustomerID	84.9%

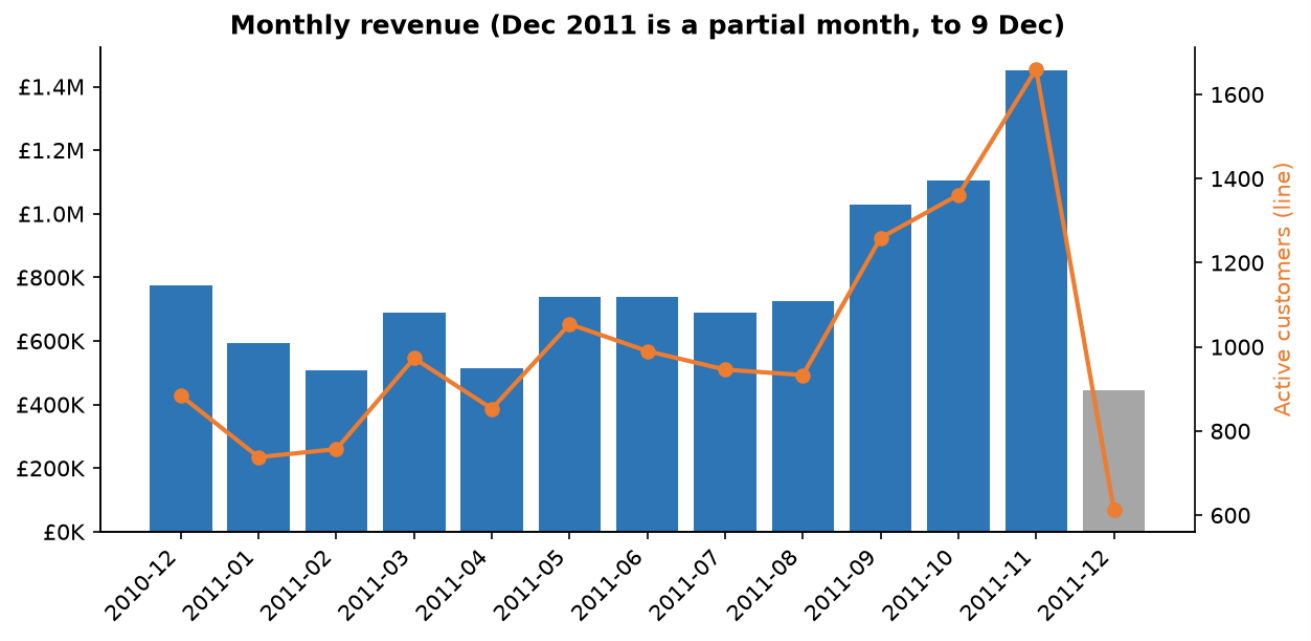


Figure 1: Monthly revenue and active customers. December 2011 is partial and ends on 9 December.

BUSINESS QUESTION AND METHOD

The source is the Online Retail dataset from the UCI Machine Learning Repository, covering 1 December 2010 to 9 December 2011. The grain is one product line on one invoice. Revenue is `Quantity x UnitPrice`; no cost or profit field exists, so every value in this report is revenue.

- Revenue and product/country analysis uses every valid completed sale, including lines without a CustomerID.
- Customer behaviour, retention and RFM use only the 4,333 identified customers; these represent 84.9% of revenue.
- Cohort retention measures the share of each monthly first-order cohort that orders again after 1, 3 and 6 months. December 2010 is excluded from averages because the dataset starts then, and December 2011 is partial.
- RFM scores Recency, Frequency and Monetary value from 1 to 5 by quintile, then assigns six practical customer segments.

CLEANING AND PREPARATION

The raw export contains 541,909 lines. The cleaning script removes exact duplicates, separates cancellations, removes non-product charges and invalid quantity/price lines, standardises product descriptions by StockCode, and removes two very large sale/cancellation pairs that were reversed in full. The cleaning log records 15 steps and every row count.

Issue	Treatment	Rows
Exact duplicate lines	Keep first occurrence	5,268
Cancellations	Move to separate cancellation analysis	9,251
Non-product lines	Remove postage, fees, vouchers and adjustments	2,356
Invalid quantity or price	Remove non-positive normal sales	2,493
Description variants	Use modal description per StockCode	6,784
Missing CustomerID	Keep for revenue; exclude from customer behaviour	131,388

REVENUE TREND, TIMING AND MARKETS

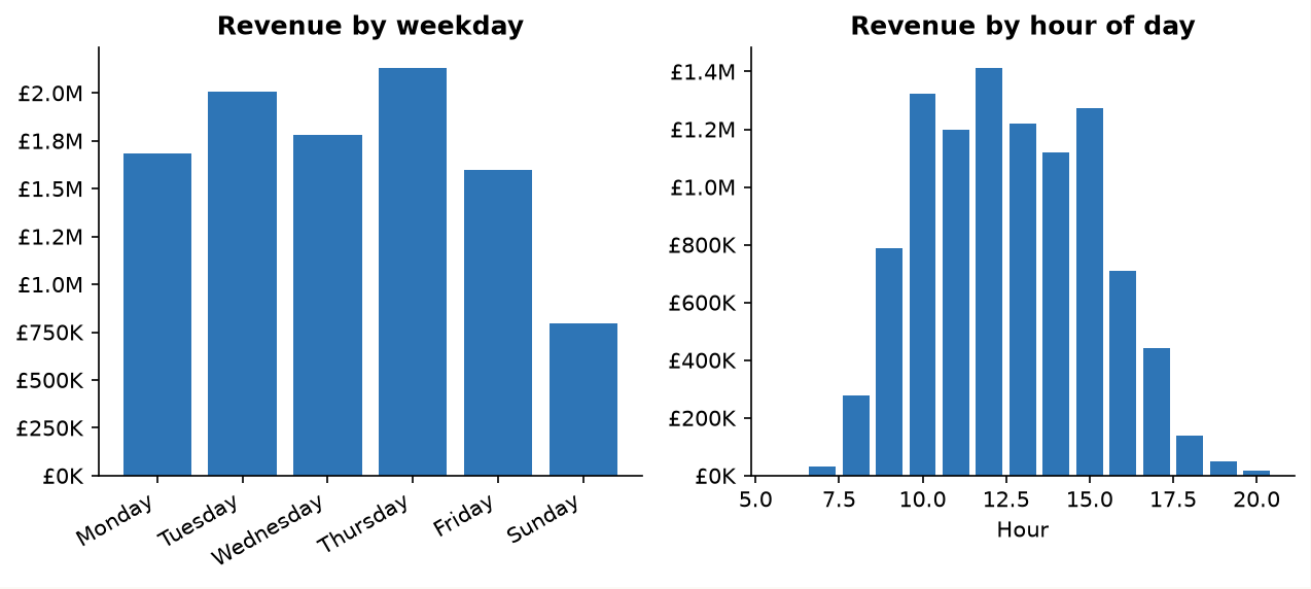


Figure 2: Revenue is concentrated in working hours and is strongest from Tuesday to Thursday.

From January to August, monthly revenue ranges from roughly £0.5M to £0.75M. It rises to £1.03M in September, £1.10M in October and £1.45M in November. September to November contributes 37.5% of full-month revenue, and November is 2.2 times the January–August average. The chart shows active customers rising with the peak, not just larger baskets.

About 75.5% of revenue is placed from 10:00 to 15:59. Thursday and Tuesday are the strongest days,

Sunday contributes 8.0%, and no Saturday orders are recorded. This is consistent with trade or business buyers, but the dataset cannot prove customer type or explain the Saturday gap.

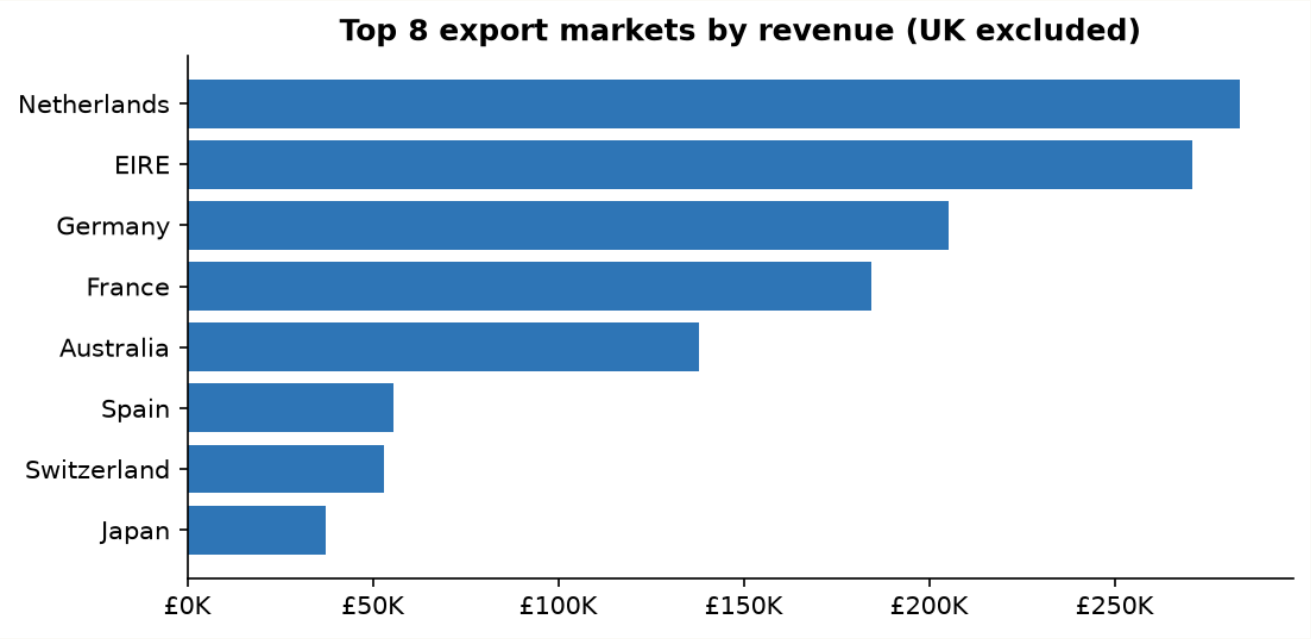


Figure 3: Top export markets by revenue, excluding the United Kingdom.

The United Kingdom produces 84.8% of revenue; exports contribute 15.2%. The largest export markets are the Netherlands (£284K), EIRE (£271K), Germany (£205K), France (£184K) and Australia (£138K). EIRE, the Netherlands and Australia are relatively concentrated in a small number of customers, while Germany and France are broader markets.

CUSTOMER CONCENTRATION AND REPEAT PURCHASE

The top 1% of customers produce 30.5% of revenue, the top 10% produce 60.3%, and the top 20% produce 73.9The mean order value is £506 while the median is £302, showing that a small number of large baskets pull up the average.

Repeat buying is the commercial engine: 65.3% of identified customers ordered more than once and repeat customers produce 93.5% of identified-customer revenue. Returning-customer baskets average £485 versus £402 for new-customer order months. The most economical growth lever is therefore turning first-time buyers into second-time buyers.

RETENTION AND NEW VERSUS RETURNING REVENUE

On average, 19.8% of a monthly new-customer cohort orders again one month later, 22.9% remains active at month three and 25.0% at month six. Retention does not decline monotonically because customers reorder in bursts, likely around their own restocking cycles. The December 2010 cohort is not a true new-customer cohort and is excluded from the averages.

Returning customers supply 79.7% of identified-customer revenue in 2011, and their contribution grows through the year. The November peak is therefore largely existing-customer expansion rather than only acquisition.

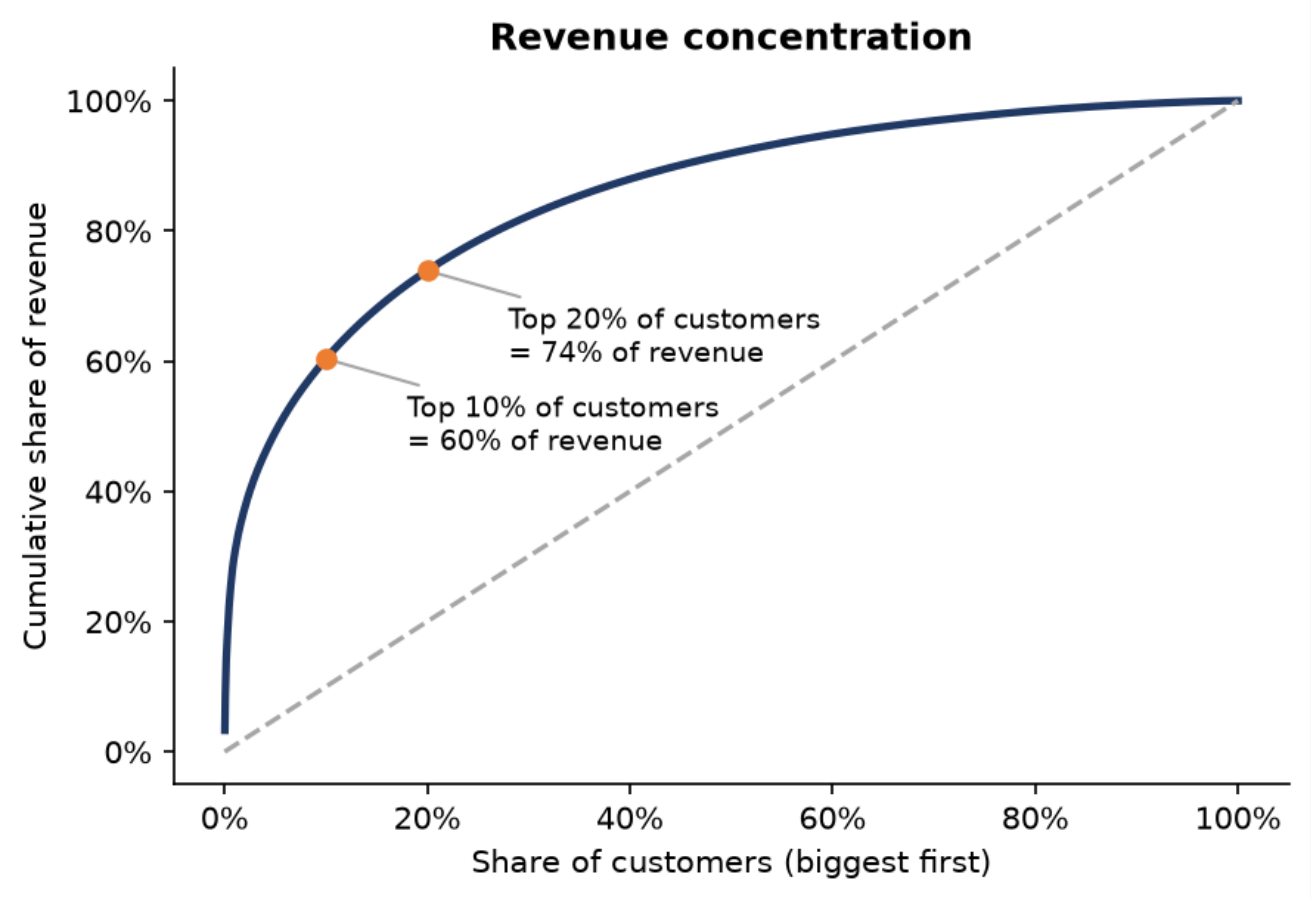


Figure 4: Cumulative revenue concentration. The top 10% of customers produce 60% of revenue and the top 20% produce 74%.

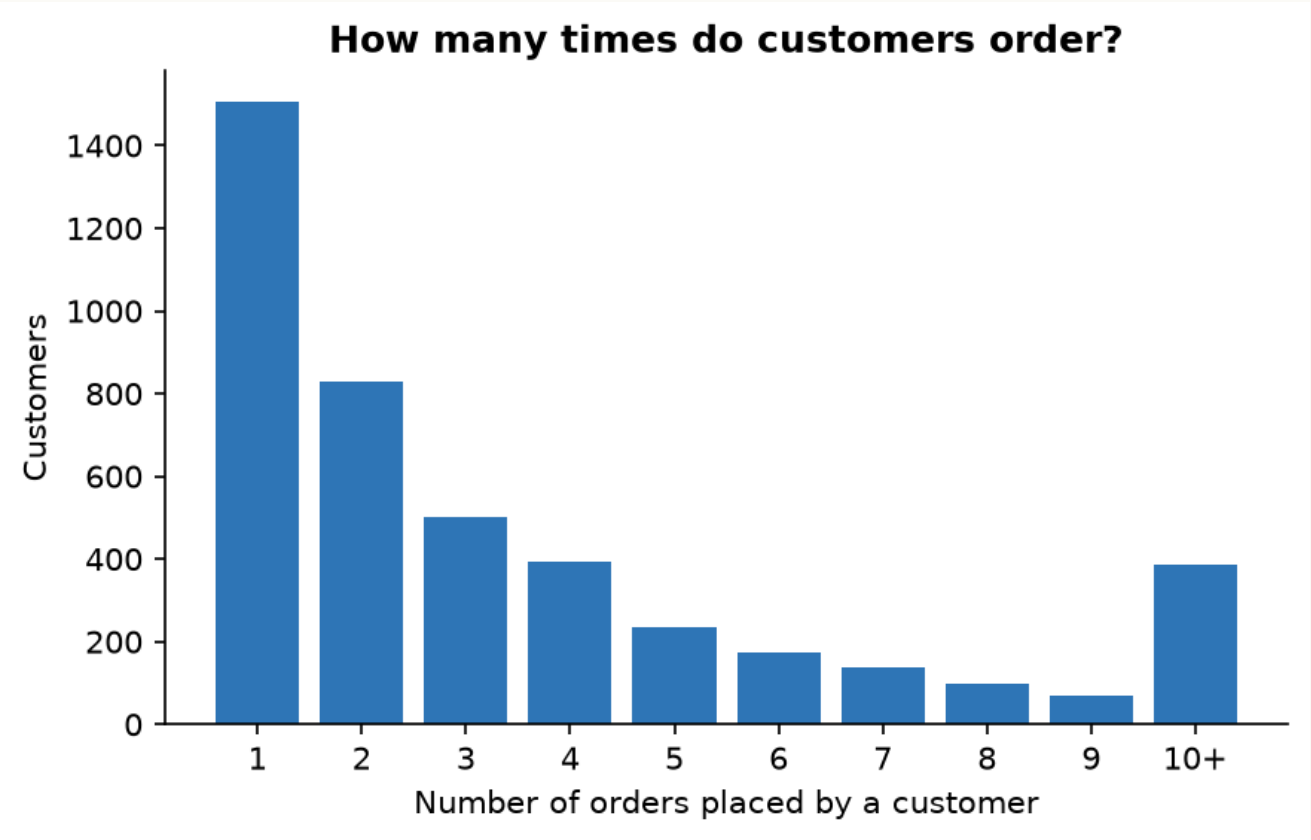


Figure 5: Distribution of order counts per identified customer.

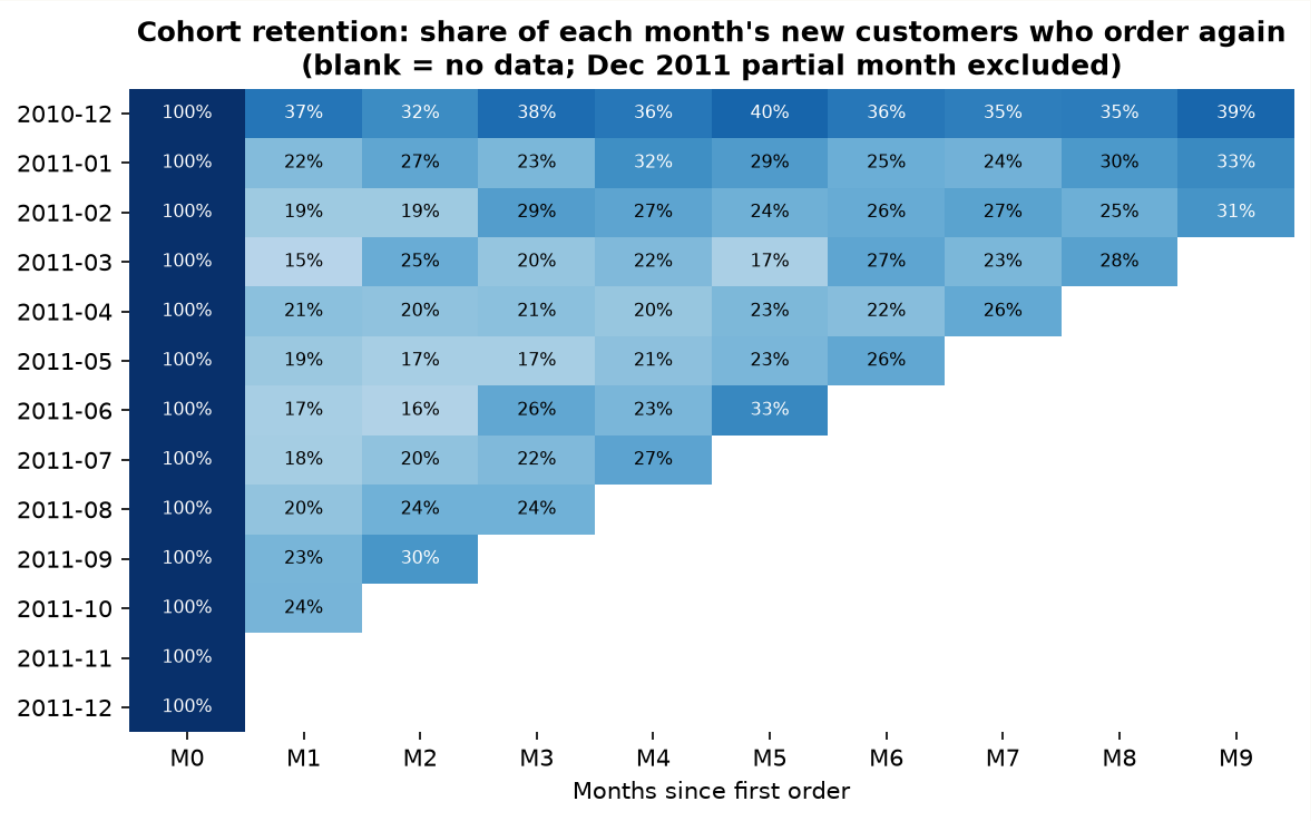


Figure 6: Cohort retention heatmap. Blank cells are unavailable or excluded because December 2011 is partial.

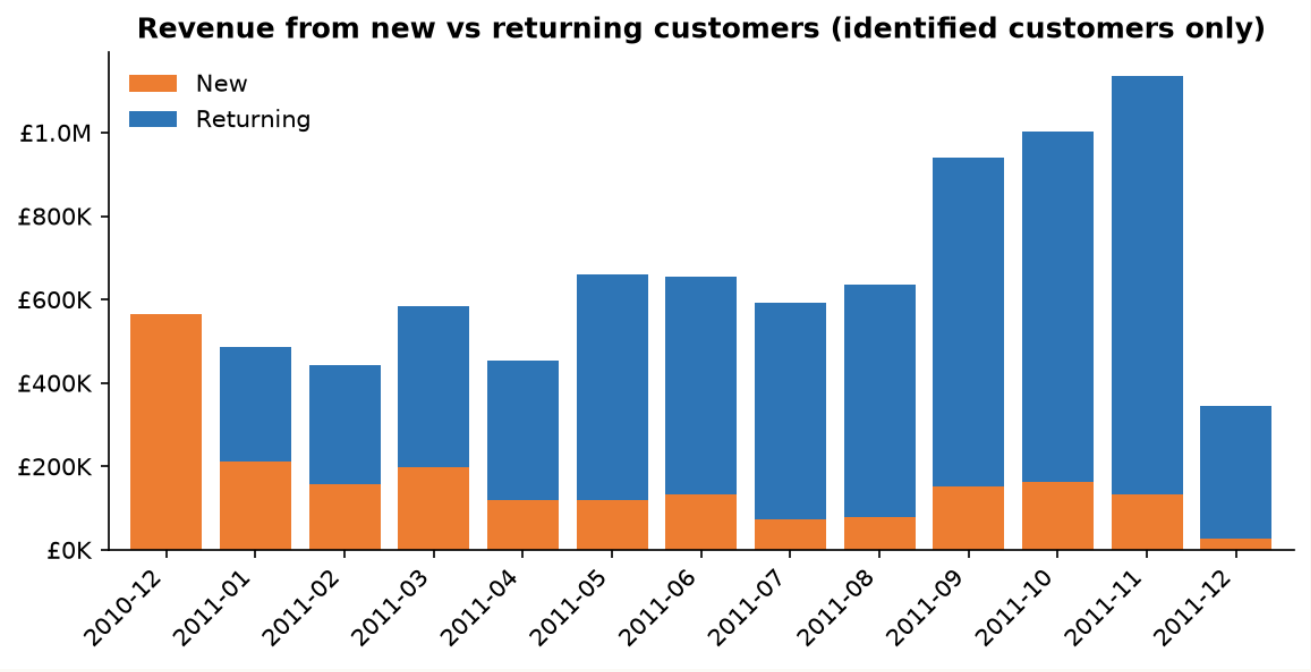


Figure 7: Monthly revenue from new and returning identified customers.

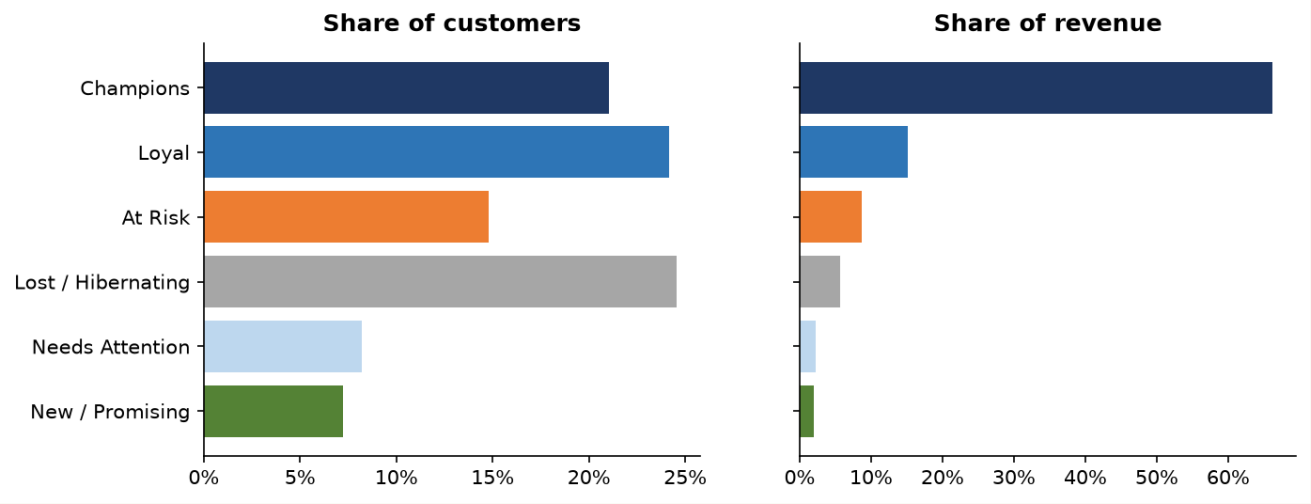


Figure 8: RFM segment shares of customers and revenue.

RFM CUSTOMER SEGMENTS

Segment	Customers	Customer share	Revenue share	Action
Champions	912	21.0%	66.2%	Protect with account ownership and early warning.
Loyal	1,047	24.2%	15.2%	Encourage frequency and broader baskets.
At Risk	641	14.8%	8.6%	Personal win-back before discounting.
Lost / Hibernating	1,064	24.6%	5.7%	Low-cost reactivation tests.
Needs Attention	356	8.2%	2.3%	Nurture and observe.
New / Promising	313	7.2%	2.0%	Drive the second order.

Champions average 11.4 orders and £6,165 in annual revenue. At Risk customers average 154 days since their last order and represent £734K of historic revenue. A narrower high-value quiet cohort contains 246 customers and £535K of prior revenue, making it a practical first win-back list.

PRODUCTS AND CANCELLATIONS

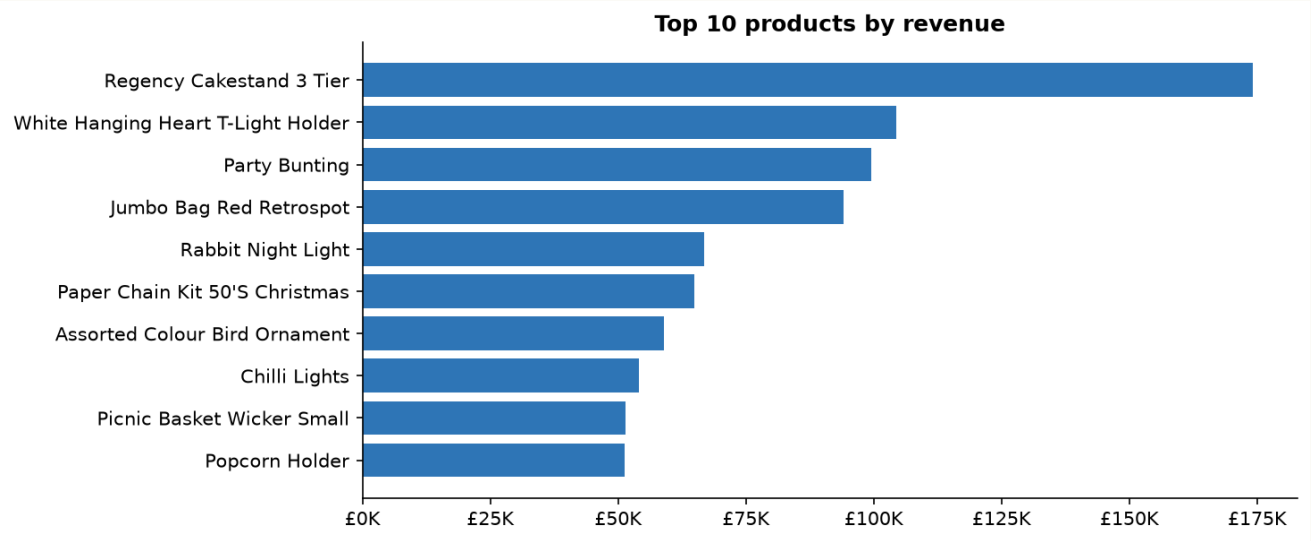


Figure 9: Top ten products by revenue. The catalogue is broad: the top ten products account for 8.2% of revenue, while the top 10% of products account for 61.0%.

The leading product is Regency Cakestand 3 Tier at £174K, followed by White Hanging Heart T-Light Holder and Party Bunting. The top ten products represent only 8.2% of revenue, so availability matters

across a wide catalogue rather than for one dominant item.

Cancellations total £230K, or 2.3% of revenue. April 2011 is the peak month at 6.5%, and the ten customers with the most cancelled value account for 34.2% of cancellation value. Cancellations are worth monitoring but are not the first growth priority.

RECOMMENDATIONS AND MEASUREMENT

1. **Protect Champions.** Assign account owners to the top 10% and alert the team when a customer's gap exceeds their normal ordering interval. Measure Champion retention and order frequency.
2. **Win back lapsed high-value customers.** Contact At Risk and valuable quiet customers personally before offering discounts. Test outreach against a holdout group and measure reactivation revenue.
3. **Get the second order.** Send a follow-up two to three weeks after the first purchase with relevant best sellers and a modest incentive. Measure second-order conversion within 90 days.
4. **Build Q4 from August.** Forecast products, stock and staffing before September, then compare Q4 revenue, stock-outs and service levels with plan.
5. **Time communications.** Prioritise Tuesday–Thursday mornings and maintain support coverage from 10:00 to 16:00. Measure conversion by send time.
6. **Use different export strategies.** Treat EIRE, the Netherlands and Australia as key-account markets; use broader market marketing in Germany and France.

LIMITATIONS

This is one retailer over one year, so seasonality cannot be separated from growth. Around 25% of lines have no CustomerID, and customer findings cover identified customers only. There is no cost, profit, marketing, customer-type or price data, so explanations are hypotheses rather than causal conclusions. RFM boundaries are quintile-based and intentionally pragmatic. Cancellations are analysed as revenue leakage, but the dataset does not provide a reason for each cancellation.

REPRODUCIBILITY

```
cd "d:\ShadowFox Intership\shadowfox-data-analyst\02-intermediate"
python scripts\clean_online_retail.py
python analysis\analyze_online_retail.py
python analysis\build_report.py
pdflatex -interaction=nonstopmode -halt-on-error -output-directory report report\on
```

The Excel analysis tables, nine chart PNGs, Word report and this PDF are all generated from the cleaned CSV outputs.